

X.509 Digital Certificate - Exam Revision Notes

Definition

X.509 is a standard digital certificate format defined by the International Telecommunication Union (ITU). It serves as a digital passport for authentication and secure communication over networks.

Purpose

- **Identity Verification:** Confirms the identity of a user, device, or server.
- **Key Distribution:** Provides the public key necessary for encryption and digital signatures.
- **Security:** Ensures data privacy and secure transactions.

Issuance

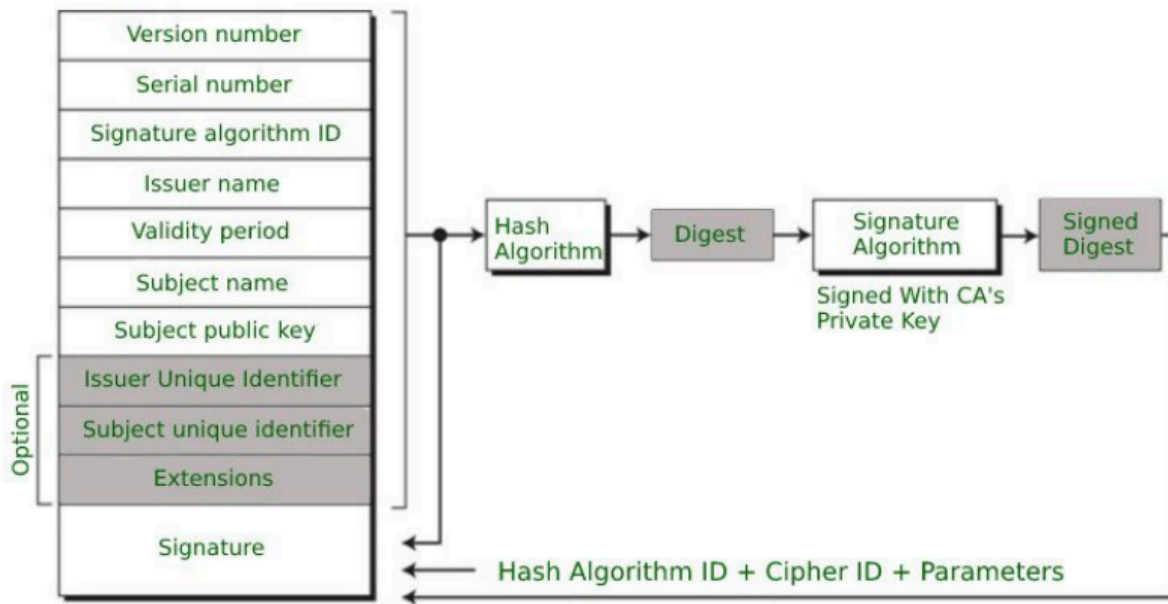
- **Certificate Authority (CA):** Certificates are issued by a trusted CA.
- **Authentication:** The CA signs the certificate with its own **private key** to guarantee its authenticity.

Structure and Fields

The X.509 certificate contains specific data fields to ensure its validity:

- **Version:** The X.509 version number (e.g., v1, v2, or v3).
- **Serial Number:** A unique identifier assigned by the CA.
- **Signature Algorithm ID:** The algorithm used to sign the certificate.
- **Issuer Name:** The name of the CA that issued the certificate.
- **Validity Period:** The "start" and "expiration" dates.
- **Subject Name:** The name of the entity (owner) the certificate belongs to.
- **Subject Public Key:** The actual public key of the owner.
- **Extensions:** Optional additional information (used in v3).
- **Digital Signature:** The hash of the certificate data encrypted by the CA's private key.

Format of X.509 Authentication Service Certificate:



How It Works

1. **Creation:** The CA compiles the owner's information and public key.
2. **Hashing:** A hash of this certificate data is generated.
3. **Signing:** The CA encrypts the hash using its **private key** to create the digital signature.
4. **Verification:** The receiver uses the **CA's public key** to decrypt the signature and verify that the data has not been tampered with.

Applications

- **SSL/TLS:** Enabling HTTPS for secure website browsing.
- **Email Security:** Protecting communications (S/MIME).
- **Code Signing:** Verifying the integrity of software downloads.
- **SSH Authentication:** Secure remote login.

Public Key Infrastructure (PKI)

PKI is the framework used to manage, store, and distribute certificates. It includes:

- **CA (Certificate Authority):** Issues and signs certificates.
 - **RA (Registration Authority):** Verifies the identity of entities before the CA issues the certificate.
 - **Certificate Database:** Stores and manages certificates.
-

Exam Tip

Remember the formula: **X.509 = Identity + Public Key + CA Signature**